

JKBOSE

2026 CLASS 11TH

Part

*→ Motion in 2D ✓
motion in plane ✓*

VECTOR

Maths ✓

LECTURE-1

PHYSICS

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Topics To Be Covered

- 1** Revision (10)
- 2** Vector ✓
- 3** Types of the Vector ✓
- 4** Algebra of Vector ✓
- 5** Triangular Law of Vector Addition ✓

~~#~~ Work Hand

~~#~~ D

”

Garwalon ne tumhe naam de diye,

Ab pehchaan tumhe banaani hai.”

Recap From 1st Lec.

Motion

- move one place to another place with respect to

Rest

Donot move from one place to another place with respect to

Reference point
(Motion is Relative)

2020

Numericals

$$1 \frac{\text{km}}{\text{hr}} = \frac{5}{18} \text{ m/s}$$

distance	displacement
• Total path • metre • <u>Scalar</u> $\times$ • Positive	• Shortest path • metre • <u>vector</u> $\vec{x}$ • + - zero
Speed	Velocity
• Speed = $\frac{\text{distance}}{\text{time}}$ • Scalar • m/s tive	• Velocity = $\frac{\text{displacement}}{\text{time}}$ • <u>Vector</u> $\frac{\text{m}}{\text{s}}$ • tive -ive 0

Recap 2nd lecture



Speed

no direction
(scalar)

Average Speed ✓

$$AS = \frac{\Delta x}{\Delta t}$$

Instantaneous Speed

$$IS = \frac{dx}{dt}$$

Velocity

direction
(vector)

Average Velocity

$$AV = \frac{\Delta \vec{x}}{\Delta t}$$

Instantaneous velocity

$$IV = \frac{d\vec{x}}{dt}$$

Lec. 3rd Recap.

Acceleration

⇒ Rate of change of velocity

$$a = \frac{\Delta v}{t} \quad \# \text{ vector} \quad \# \text{ m/s}^2$$

⇒ $a = +ve$, $a = 0$, $a = -ve$

Retardation ♡

Motion

Uniform Motion

$$a = 0$$

Non-uniform Motion

$$a > 0$$

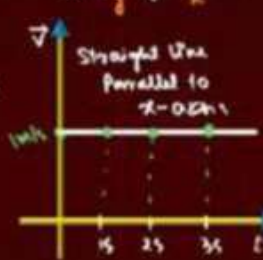
$$a < 0$$

Graphs

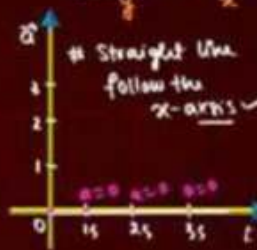
① displacement-time graph



② velocity-time

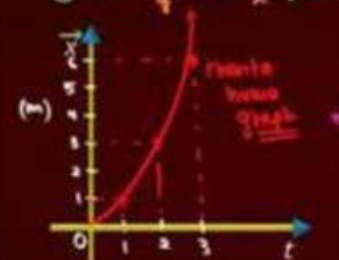


acceleration-time



Graphs

① displacement-time graph



② velocity-time



acceleration-time

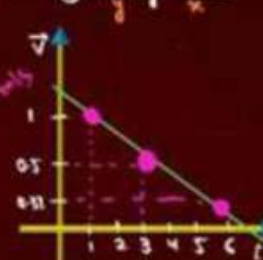


Graphs

① displacement-time graph



② velocity-time



acceleration-time

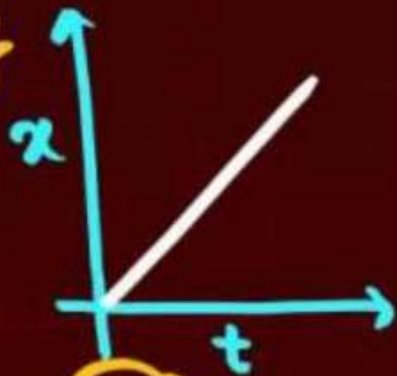


Recap From 4th Lec

→ Board

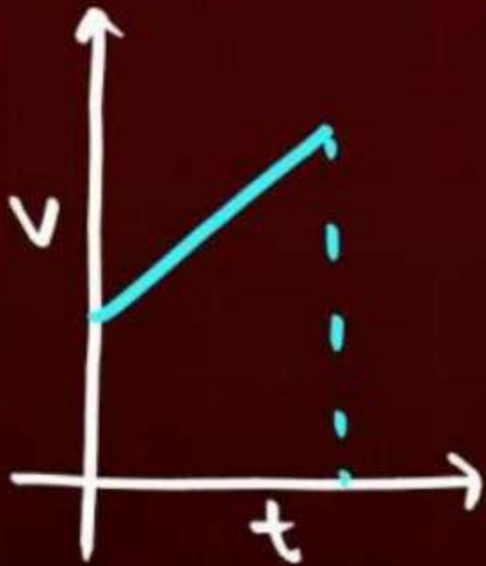
✓ Slope of $(x)t$ graph gives

Velocity

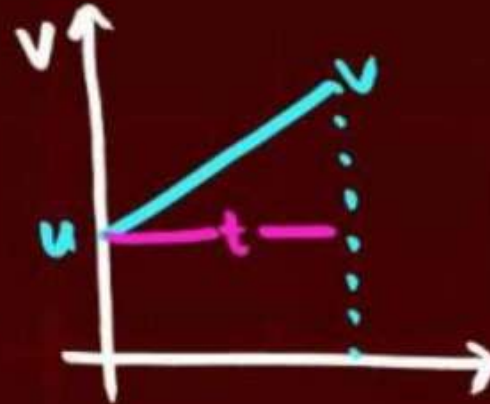


$$v = \frac{dx}{dt}$$

✓ Area of $(v)t$ graph gives
displacement = displacement(x)



Newton equation of motion



$$\textcircled{1} v = u + at$$

$$\textcircled{2} s = ut + \frac{1}{2}at^2$$

$$\textcircled{3} v^2 = u^2 + 2as$$

Acceleration due to gravity (g)

Ⓜ

↓ g

$$g = -10 \text{ m/s}^2$$

$$g = -9.8 \text{ m/s}^2$$



$$v = u - gt$$

$$s = ut - \frac{1}{2}gt^2$$

$$v^2 = u^2 - 2gs$$

lecture . no . ①

$$p \propto n u$$

Physical Quantity $\rightarrow$ Joo App measure

10th

Scalar pQ

Direction

$\rightarrow$ Kitna hai.

It has only magnitude

Mass = 40 kg

Length = 50 m

Time = 1 hr

Vector pQ

It is the pQ which have magnitude + Direction

Kitni hai

$\Rightarrow$ displacement $\vec{x}$

$\Rightarrow$ velocity $\vec{v}$

$\Rightarrow$ acceleration $\vec{a}$

Force $\vec{F}$
momentum $\vec{p}$

Feel of vector

6 - Saal Switzerland (M)

Kashmir (Bandipora)
20km

Bandipora

20km

lal Bazar

20km

Wm

la' chok

Jammu

lal Bazar = 100km
Tammu = 180km
Khistwar = 60km
Bandipora = 20km

khistawar

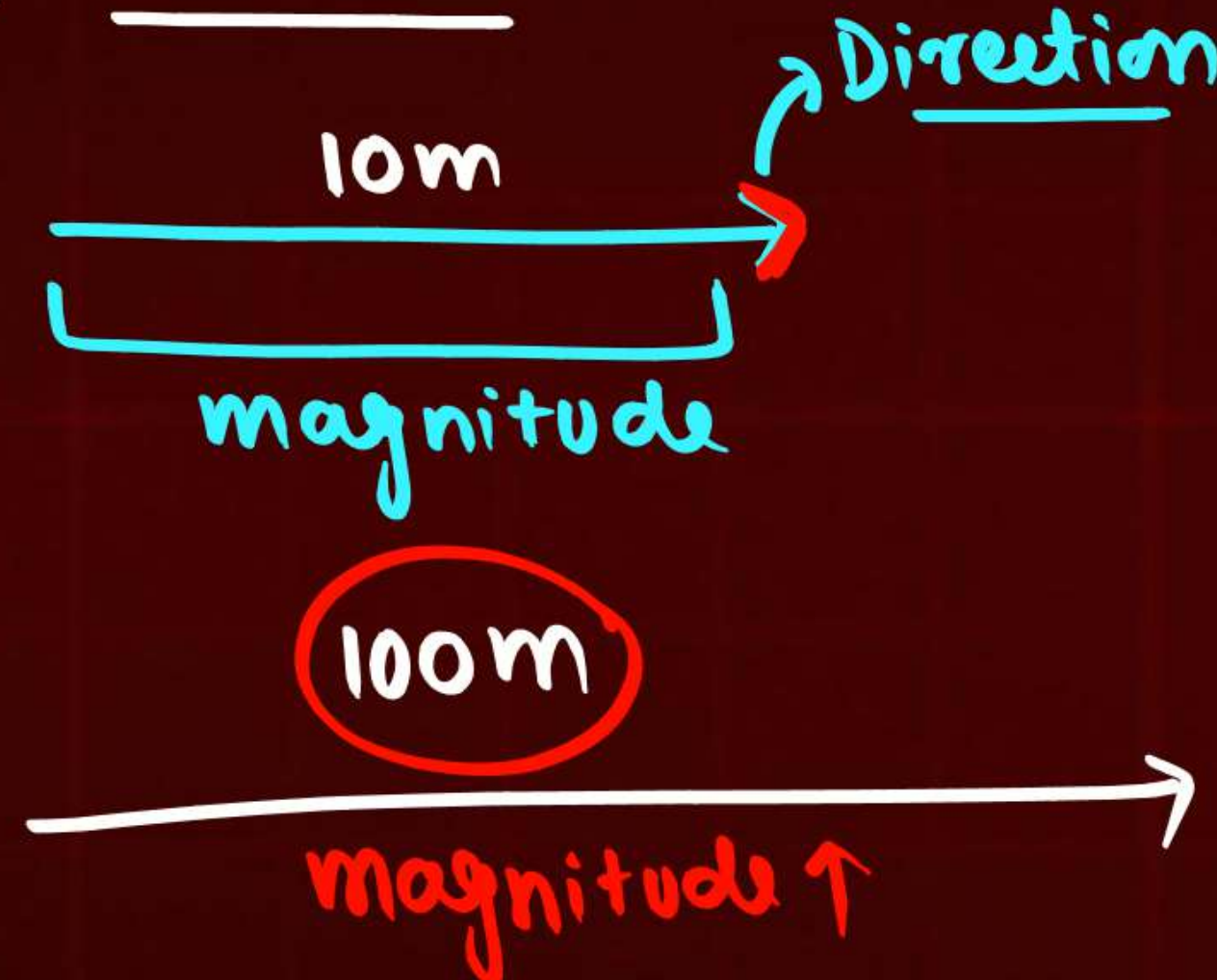
Vector

The vector is the PQ which have magnitude & direction

- Displacement $\vec{x}$
- Velocity $\vec{v}$
- Momentum $\vec{p}$
- Force $\vec{F}$
- Acceleration $\vec{a}$



Representation of vector

- ① Arrow
- 10m
- Direction
- magnitude
- 100m
- magnitude ↑
- 

- ② Bold letter
- A → **A**

- ③ letter with arrow on a head.

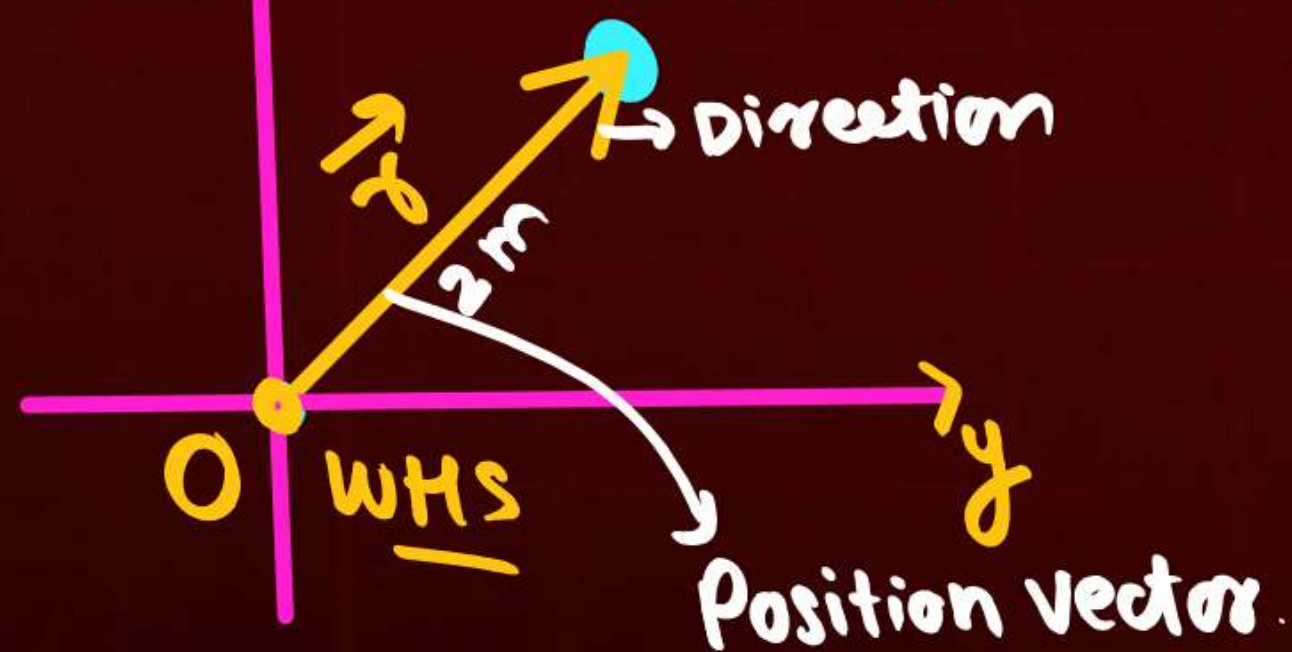
$\vec{x}$ $\vec{a}$

$\vec{v}$ $\vec{f}$

Types of the Vector

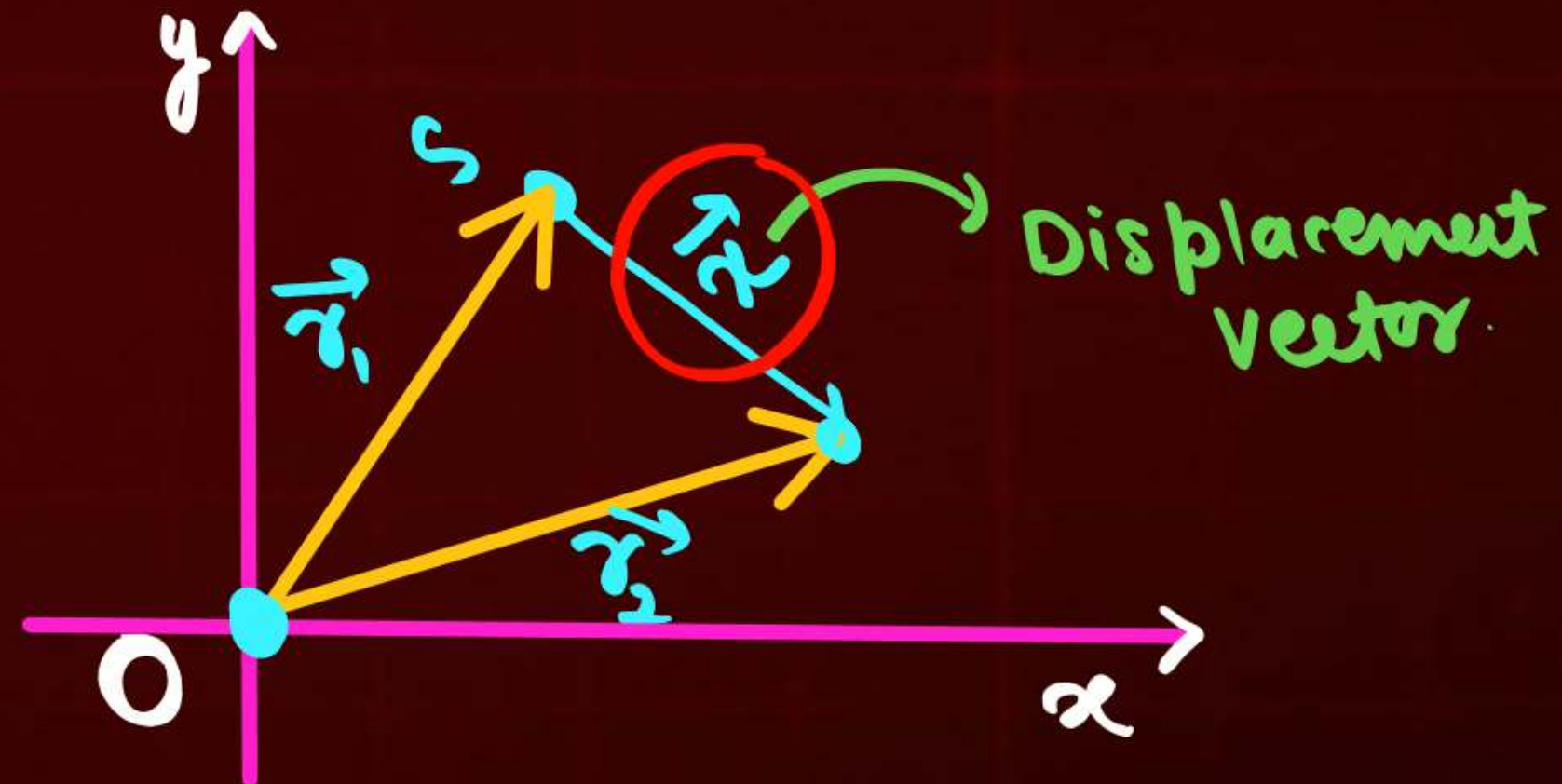
1. Position Vector

=> The vector which tells the position of Particle/object



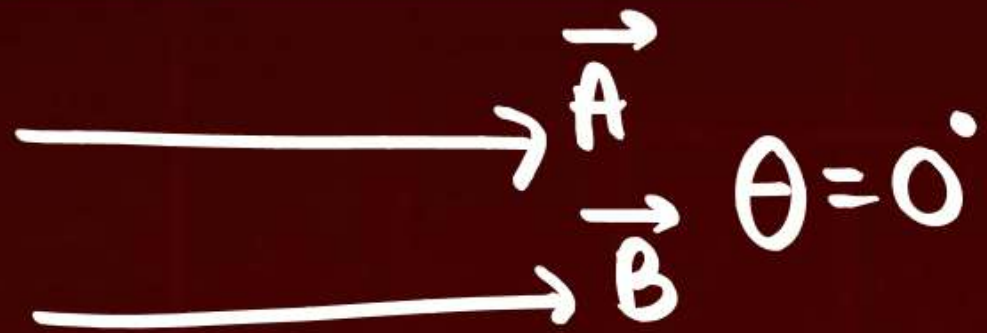
2. Displacement Vector

the vector which shows the displacement.



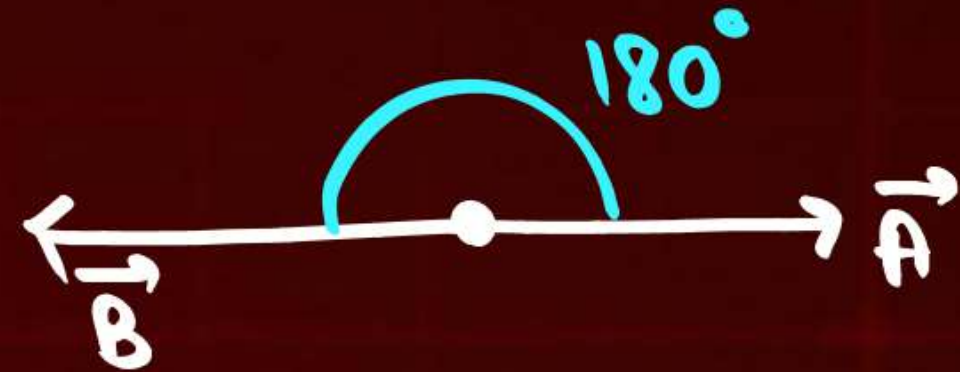
Types of the Vector

3. Parallel Vector



When two vector
have 0° angle b/w
them

4. Anti Parallel Vector



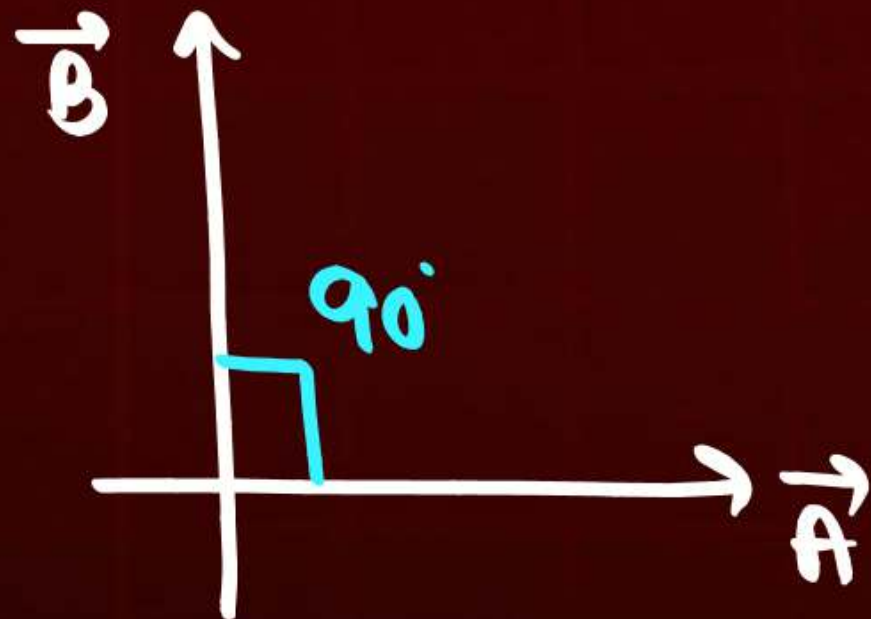
When two vectors are in
opposite direction.

Types of the Vector



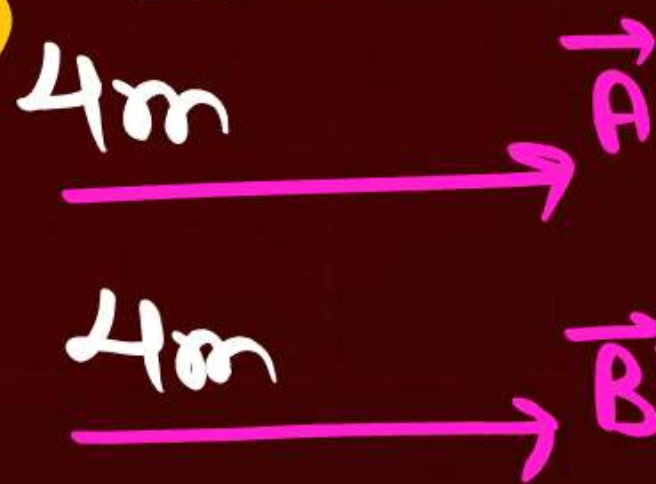
5. Orthogonal Vector

When two vectors are perpendicular to each other



6. Equal Vector

When two vectors have equal magnitude and equal direction.



$$\vec{A} = \vec{B}$$

Parallel

Types of the Vector

7. Unit Vector (Sab Se Zada use Hone walla vector)

① • The vector which have magnitude = ①

$$\text{unit vector} = \underline{m} \times D$$

$$\text{unit vector} = 1 \times D$$

$$\text{unit vector} = \text{Direction}$$

② It only give direction

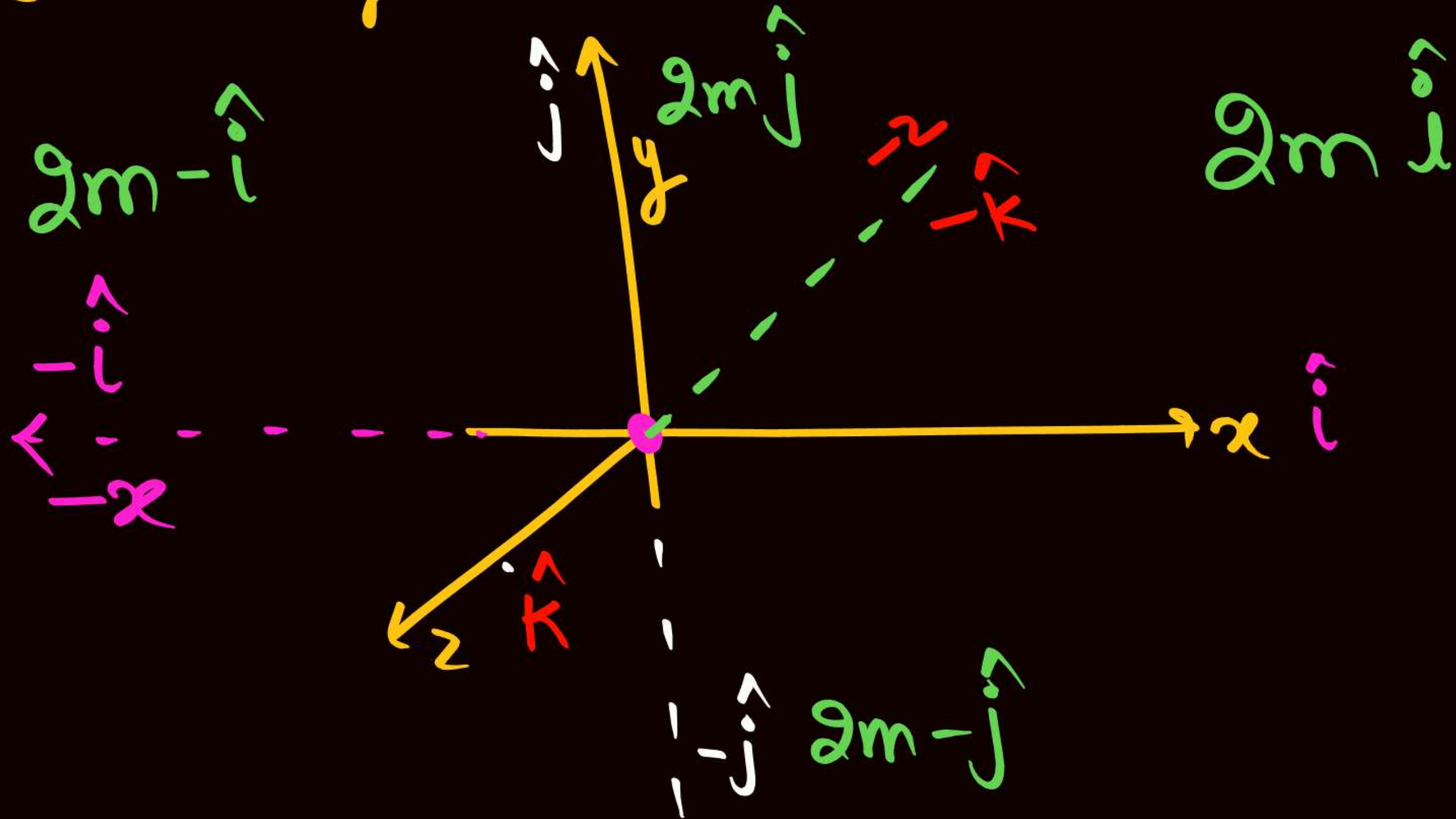
③ It is Represented By $\hat{A}$

④ Formula

$$\hat{A} = \frac{\vec{A}}{A} \rightarrow \text{vector } A \text{ magnitude of } A$$

$$\hat{A} = \frac{\vec{A}}{A} = \rightarrow$$

⑤ Commonly 3 direction hi use hoti hai.



Algebra of Vector



Maths of vector

$\oplus$ Add

$\ominus$ Subtract

$\otimes$ Multiple

$\div$ Divide

$(\vec{A} + \vec{B})$

Add

angle



$$F_1 = 2N$$

$$F_2 = 2N$$

$$F_{net} = \underline{4N}$$

Q



Sol

$$f_{net} = 2 - 2 = 0$$

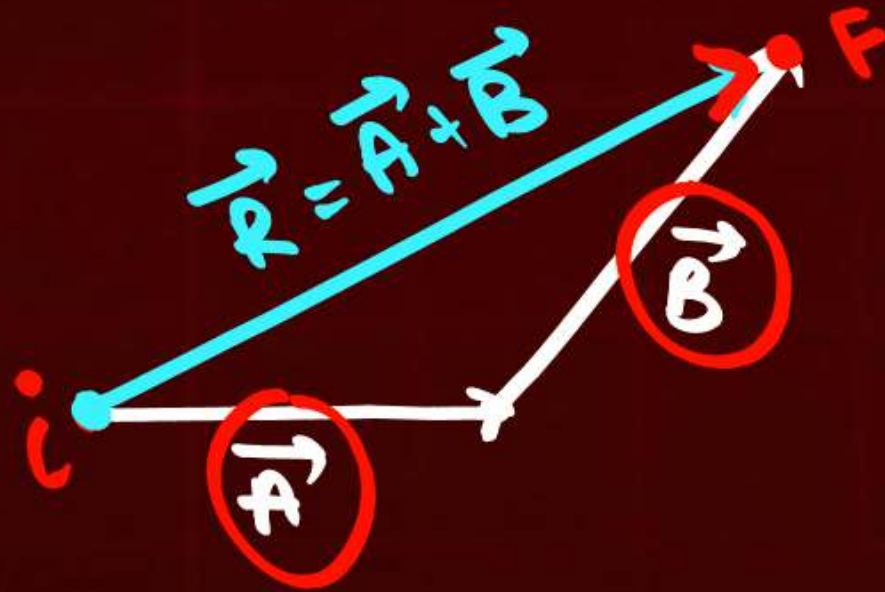
Do vector ko (+, -, $\times$ and $\div$) Karne
Sa phella

Thumhai unke beeh ka
Angle pata hona chahiya.

Maths of Vector

1. Addition of Vector (How to add the two vector)

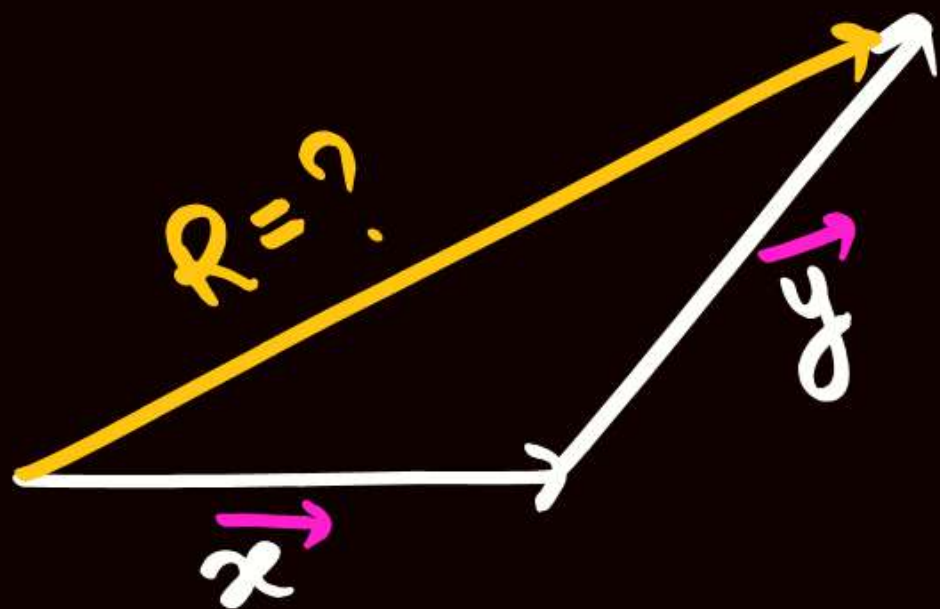
a. Triangular Law of Vector addition — 2019 — 2020.



If two vector
are the Adjacent
Side of triangle Δ
then third Side is

the Resultant vector in opposite
sense.

11Q

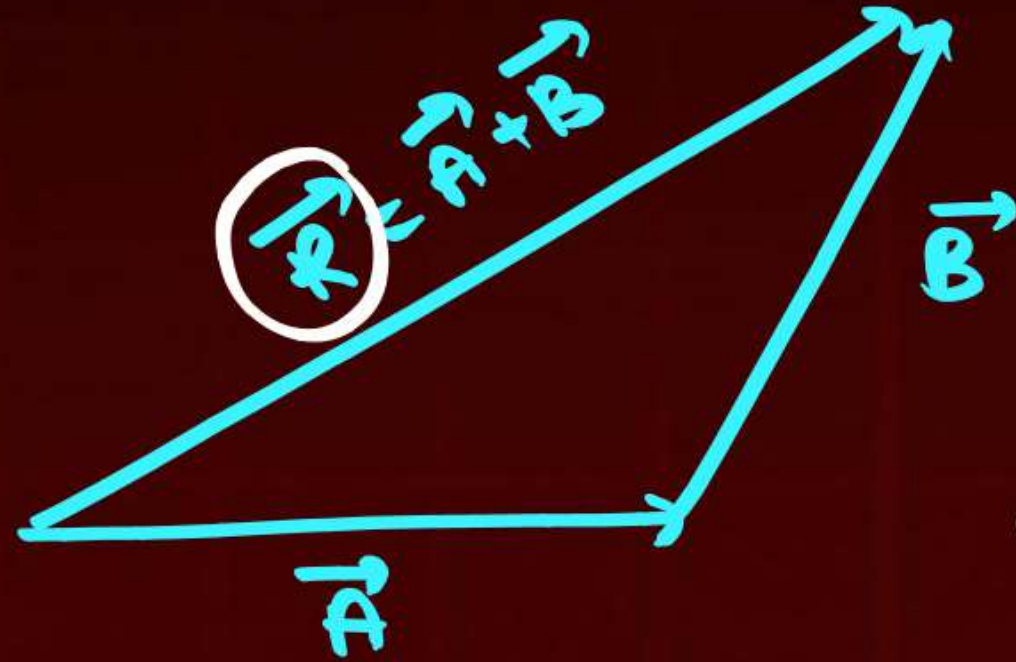


~~(a)~~ $\vec{x} + \vec{y}$

(b) $\vec{x} - \vec{y}$

(c) $R = \vec{x}$

(d) $R = \vec{y}$



② 119m law of
Vector Addition

$$R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$$

Next
class.

SUMMARY

Vector ✓

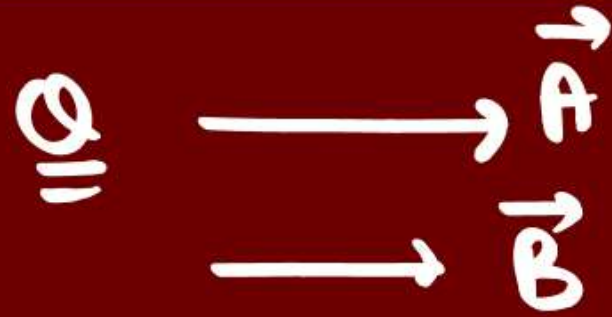
Types of Vector

Triangular law of VA

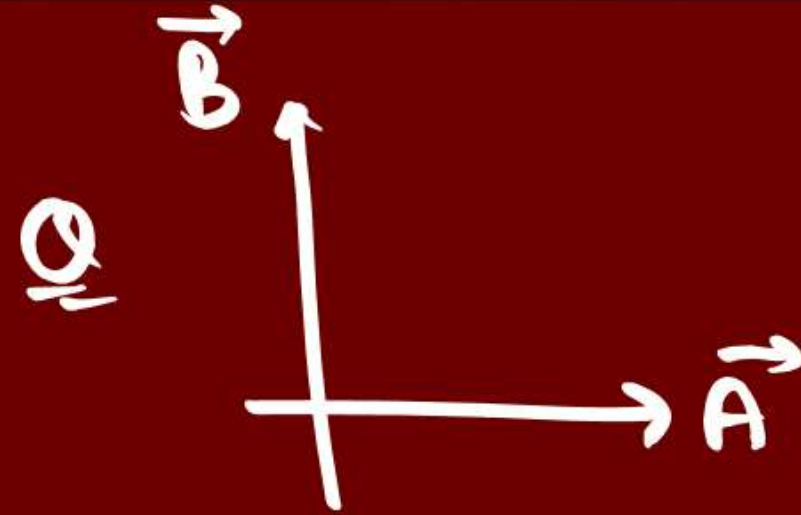
$$\# R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$$

↓
Next class

HOMework



_____ vector
vector



_____ vector

Thank
You